Subject: Bio201 (Past papers)

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Q Incomplete Dominance
Definition: Incomplete dominance is a form of intermediate inheritance in which one allele for a specific trait is not completely expressed over its paired allele. This results in a third phenotype in which the expressed physical trait is a combination of the phenotypes of both alleles.

Q Define Codominance?

It is a form of dominance wherein the alleles of a gene pair in a heterozygote are fully expressed. This results in offspring with a phenotype that is neither dominant nor recessive. A typical example showing codominance is the ABO

Q……energy is the energy of motion
(a) Kinetic
(b) Potential
(c) Chemical
(d) Atomic
Q
Ribosomes can be free or attached with
(a) golgi apparatus
(b) lysosomes
(c) endoplasmic reticulum
(d) necleus
Q
Which organ/molecule can convert light energy
(a) NAD
(b) Retina
(c) Biotin
(d) Coenzyme A
Q
Beta pleated sheet is found in
(a) Primary protein structure
(b) Secondary protein structure
(c) Tertiary protein structure
(d) Quaternary protein structure
Q
Inhibitor that binds to a site other than active site of enzyme

(a) Competitive inhibitor
(b) Irreversible inhibitor
(c) Reversible inhibitor
(d) Non competitive inhibitor
Q
lipid act as a following in which many components of cell membrane floats
(a) Pond
(b) Lake
(c) River
(d) Stream
Q
How many macromolecules a cell has....
(a) 2
(b) 4
(c) 6
(d) 8
Q
Beta pleated sheet is found in
(a) Secondary protein structure
(b) Primary protein structure
(c) Tertiary protein structure
(d) Quaternary protein structure
Q
Cell membrane expels particles by vesicle formation
(a) pinocytosis
(b) Phagocytosis
(c) Endocytosis
(d) Exocytosis
Q
Molecule of phospholipids has ----------
(a) hydrophobic & Hydrophilic domain
(b) hydrophobic only
(c) hydrophilic only
(d) hydrogen linkage
Q
If one gene prevent the expression of other gene
(a) Pleiotropy
(b) Epistasis
(c) Synapse
(d) Tetrad

Q
Phenylalanine hydroxylase metabolizes
(a) Phenylalanine to Lysine
(b) Phenylalanine to Tyrosine
(c) Tyrosine to Phenylalanine
(d) Lysine to Phenylalanine
Q
Restriction enzymes inhibit the growth of
(a) Bacteriophage
(b) Bacteria
(c) Algae
(d) None
Q:
Missing of enzyme homogentisic acid oxidase caused
(a) Phenylketonuria
(b) Parkinson disease
(c) Alkaptonuria
(d) Cystic fibrosis
Q
Microtubules are originated from
(a) Cytoskeletal
(b) Nucleus
(c) Proteins
(d) Centrosomes
Q
The disintegration and condensation of nuclear material takes place in
(a) prophase II
(b) anaphase II
(c) metaphase II
(d) telophase II
Q
RNAs in large subunit of Eukaryotic ribosome
(a) 1
(b) 2
(c) 3
(d) 4
Q
Centromeres separate during whose anaphase:
(a) Mitosis
(b) meiosis I

(c) meiosis II
(d) None
Q
Enormous genetic information is present in
(a) RNA
(b) DNA.
(c) Nucleolus
(d) Nucleoplasm
Q
Okazaki fragments are joined by
(a) Ligase
(b) Gyrase
(c) Helicase
(d) Primase
Q
How much water present in a cell.....
(a) 50%
(b) 60%
(c) 70%
(d) 80%
Q
Charge on DNA is
(a) Positive
(b) Negative
(c) Neutral
(d) None
Q
Which of the following stabilizes cell shape
(a) DNA
(b) RNA
(c) Actin
(d) Myosin
Q
Define following terms?
(a) Endocytosis
(b) Phagocytosis
(c) Pinocytosis
(d) Exocytosis
Q
Which of the following stabilizes cell shape

(a) DNA
(b) RNA
(c) Actin
(d) Myosin
Q
……energy is the energy of motion
(a) Kinetic
(b) Potential
(c) Chemical
(d) Atomic
Q
Space between double membrane structure of endoplasmic reticulum is called
(a) Vesicles
(b) Leumen
(c) Cavity
(d) endoplasmic space
Q
----------is one of the example of co-factor
(a) Copper
(b) Biotin
(c) Haeme
(d) All
Q
lipid act as a following in which many components of cell membrane floats
(a) Pond
(b) Lake
(c) River
(d) Stream
Q
DNA replicates during.....
(a) G0 phase.
(b) G1 phase
(c) G2 phase
(d) S phase
Q
Difference between the alpha 1-4 glycosidic and beta 1-4 glycosidic bonds is on the basis of :
(a) Specificity
(b) Stability

(c) neutral pH
(d) size and shape
Q
Example of variable expressivity
(a) Dementia
(b) Huntington disease
(c) Alzheimer disease.
(d) Pancreatic necrosis
Q
Following has a major role in down syndrome
(a) Maternal age
(b) Fraternal age
(c) Child age
(d) Relative age
Q
DNA replication obeys
(a) Dispersive model
(b) Conservative model
(c) Semi-conservative model
(d) Distractive model
Q
Missing of enzyme homogentisic acid oxidase caused
(a) Phenylketonuria
(b) Parkinson disease
(c) Alkaptonuria
(d) Cystic fibrosis
Q
The RNA is --------- while DNA is ----------
(a) double stranded, double stranded
(b) single stranded, double stranded
(c) double stranded, single stranded
(d) partially double stranded, double stranded
Q
Ribosome after protein formation arrives at
(a) Lysosome
(b) Rough endoplasmic reticulum
(c) Smooth endoplasmic reticulum
(d) Golgi bodies
Q

The nucleus membrane is continuous with
(a) golgi apparatus
(b) ribosomes
(c) endoplasmic reticulum
(d) nucleus
Q
Power generating organelle of cell is:
(a) golgi bodies
(b) nucleus
(c) mitochondria
(d) cell membrane
Q
The disintegration and condensation of nuclear material takes place in
(a) prophase II
(b) anaphase II
(c) metaphase II
(d) telophase II
Q
An observable feature in the subject of genetics is called
(a) Trait
(b) Character
(c) Phenotype
(d) All
Q
Ribosome after protein formation arrives at
(a) Lysosome
(b) Rough endoplasmic reticulum

1: property of lipids (2)

Hydrophobic molecules
Water insoluble
Aggregate together (byhydrophobic and Van derWaals force)

Functions

Store energy
Phospholipidsmembranes
Carotenoids & Light
Hormones and vitamins
Animal fat is insulator
Insulation of nerves

Water repellant for skin,
Fur and feathers

2: differencepurine& pyrimidine (2)

Purines and Pyrimides are the nitrogenous bases are make the two different nucleotides bases in DNA and RNA.The two carbon nitrogen ring bases are the Adenine guanine And the Single ring nitrogen carbon bases are Thiamin and cytosine

3: note on transcription (5)

Initiation begins at promoter -tightly binds RNA Pol
1) Where to start transcription.
2) Which strand to read?
3) The direction to take start from.
Part of promoter is initiation site (transcription start).

RNA Pol moves 3' to 5', synthesis 5' to 3', primer
Unwinds 20 bp at a time.
Transcript antiparallel to DNA template strand.
No proof reading error rate 1 in 10,000-100,000

Pre-mRNA 1st product of transcription in
Eukaryotes

techniques of genetic engineering (long)

Process history choosing target genes gene manipulation extraction from cell gene isolation modification transformation development regeneration selection confirmation gene targeting gene trapping

dilemmas of DNA replication

,mitochondrial features

, restriction endonucleases

Special bacterial enzyme can cut double
stranded DNA at specific sites in a test tube.

Restriction endonucleases (restriction enzymes) act as a kind of immune system, protecting the cell from the invasion of foreign DNA (virus).
Many recombinant DNA technologies, which the field of biotechnology heavily relies on, are unlikely to have been developed without the discovery of restriction enzymes

,,macromolecules present in cell.

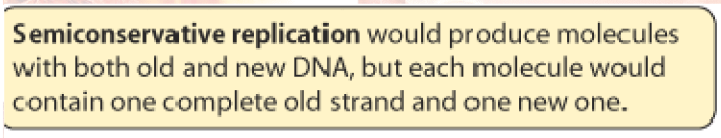
Carbohydrates

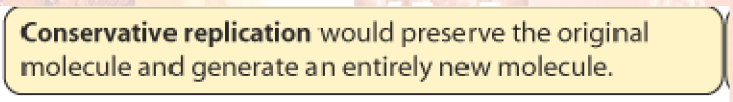
Proteins

Lipids

Nucleic acid

Q1.diffentiale between dispersive, conservative , semi-conservative model of dna(5)





Q2.difference between Nucleoside vs Nucleotide?(3)

Molecule which contains pentose sugar+Nitrogen base called Nucleoside

Molecule which contain pentose sugar + Nitrogen base + Phosphate group

Q.3.difference between dna and rna?(5)

DNA += Deoxyribonucleic acid –Double stranded molecule- contain Deoxyribose sugar

RNA = Ribo nucleic acid – single stranded molecle contain ribose sugar
Q.4.Describe process of mitosis?(5)

The process of cell division in which single cell divide into two new daughter cell which have same number of chromosomes

Mitosis process which contain different stages

Prophase

Meta phase

Ana phase

Telo phase

differtiate between DNA and RNA

3.3 Distinguishing RNA from DNA

NUCLEIC ACID	SUGAR	BASES
RNA	Ribose	Adenine Cytosine Guanine Uracil
DNA	Deoxyribose	Adenine Cytosine Guanine Thymine

differtiate between nucleotide and nucleoside

A molecule containing all these three components is called a **nucleotide**.
• While a molecule without the phosphate group is called a **nucleoside**.
Nucleotide = Nucleoside + Phophoric acid
&
Nucleoside = Nucleotide –Phoshoric acid

Gel electrophoresis

eletronegativity

downsyndrome

Arises from meiotic nondisjunction Normal Parents 95%

of time the extra chromosome comes from mother Maternal age plays a significant role (1 in 1550 for women under 20 and 1 in 25 for women over 45) 4% of patients have 14:21 translocation 1% of patients are mosaic

Single palmar fold (also known as a simian crease), Epicanthic Folds, Flat Facial Profile, IQ 25-50, Cardiac Malformations
mitosis step

Enlist the organelle of cell??

Components of a typical animal cell:

- Nucleolus.
- **Nucleus**.
- **Ribosome** (little dots)
- **Vesicle**.
- **Rough endoplasmic reticulum**.
- **Golgi apparatus** (or "Golgi body")
- **Cytoskeleton**.
- **Smooth endoplasmic reticulum**.

Function of DNA??

Deoxyribonucleic acid (DNA) is a nucleic acid that contains the genetic instructions for the development and function of living things. All known cellular life and some viruses contain DNA. The main role of DNA in the cell is the long-term **storage** of information.
What is three dilemma of DNA replication??

- Replication Basics. Replication depends on the pairing of bases between the two strands of DNA. ...
- Initiation. ...
- Elongation. ...
- Termination.

Diff b/w cofactor, coenzyme and pathetic grp and its example??

a substance (other than the substrate) whose presence is essential for the activity of an enzyme.

Coenzymes are small molecules. They cannot by themselves catalyze a reaction but they can help enzymes to do so. In technical terms, **coenzymes** are organic nonprotein molecules that bind with the protein molecule (apoenzyme) to form the active enzyme (holoenzyme).

Prosthetic groups are cofactors that bind tightly to proteins or enzymes. As if holding on for dear life, they are not easily removed. They can be organic or metal ions and are often attached to proteins by a covalent bond. The same cofactors can bind multiple different types of enzymes and may bind some enzymes loosely, as a coenzyme, and others tightly, as a prosthetic group.

What is the active site on an enzyme??

the **active site** is the region of an **enzyme**where substrate molecules bind and undergo a chemical reaction. The **active site** consists of residues that form temporary bonds with the substrate (binding **site**) and residues that catalyse a reaction of that substrate (**catalytic site**).

Why proteins denature?
Down syndrome defines?
Differentiate between purine and pyridine?
Describe the phases of mitosis?
What is the function of cellular membrane?
Translation


biotechnologists
biotechnologists
vu
remember us in your prayers
thanks
(bio_
technologist)
biotechnologists